Retrieve data with APIs from Python

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APIs (*Application Programming Interface*) are an expanding way of accessing data. Thanks to APIs, script automation is facilitated since it is longer necessary to store a file and manage its different versions, but only to query a database and let the data producer handle the updates.

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1 Introduction: What is an API?

In the previous chapters, we saw how to consume data from a file (the simplest access mode) or how to retrieve data through web scraping, a method that allows `Python` to mimic the behavior of a web browser and extract information by harvesting the HTML that a website serves.

Web scraping is a makeshift approach to accessing data. Fortunately, there are other ways to access data: data APIs. In computing, an API is a set of protocols that enables two software systems to communicate with each other. For example, the term "Pandas API" is sometimes used to indicate that `Pandas` serves as an interface between your `Python` code and a more efficient compiled language (`C`) that performs the calculations you request at the Python level. The goal of an API is to provide a simple access point to a functionality while hiding the implementation details.

In this chapter, we focus mainly on data APIs. They are simply a way to make data available: rather than allowing the user direct access to databases (often large and complex), the API invites them to formulate a query which is processed by the server hosting the database, and then returns data in response to that query.

The increased use of APIs in the context of open data strategies is one of the pillars of the 15 French ministerial roadmaps regarding the opening, circulation, and valorization of public data.

> **Note**
>
> In recent years, an official geocoding service has been established for French territory. It is free and efficiently allows addresses to be geocoded via an API. This API, known as the **National Address Database (BAN)**, has benefited from the pooling of data from various stakeholders (local authorities, postal services, IGN) as well as the expertise of contributors like Etalab. Its documentation is available at https://www.data.gouv.fr/dataservices/api-adresse-base-adresse-nationale-ban/.

A common example used to illustrate APIs is that of a restaurant. The documentation is like your menu: it lists the dishes (databases) that you can order and any optional ingredients you can choose (the parameters of your query): chicken, beef, or a vegetarian option? When you order, you don't get to see the recipe used in the kitchen to prepare your dish – you simply receive the finished product. Naturally, the more refined the dish you request (i.e. involving complex calculations on the server side), the longer it will take to arrive.

> **Illustration with the BAN API**
>
> To illustrate this, let's imagine what happens when, later in the chapter, we make requests to the BAN API.
>
> Using `Python`, we send our order to the API: addresses that are more or less complete, along with additional instructions such as the municipality code. These extra details are akin to information provided to a restaurant's server—like dietary restrictions—which personalize the recipe.
>
> Based on these instructions, the dish is prepared. Specifically, a routine is executed on Etalab's servers that searches an address repository for the one most similar to the address requested, possibly adapting based on the additional details provided. Once the kitchen has completed this preparation, the dish is sent back to the client. In this case, the "dish" consists of geographic coordinates corresponding to the best matching address.
>
> Thus, the client only needs to focus on submitting a proper query and enjoying the dish delivered. The complexity of the process is handled by the specialists who designed the API. Perhaps other specialists, such as those at Google Maps, implement a different recipe for the same dish (geographic coordinates), but they will likely offer a very similar menu. This greatly simplifies your work: you only need to change a few lines of API call code rather than overhauling a long and complex set of address identification methods.

Pedagogical Approach

After an initial presentation of the general principle of APIs, this chapter illustrates their use through `Python` via a fairly standard use case: we have a dataset that we first want to geolocate. To do this, we will ask an API to return geographic coordinates based on addresses. Later, we will retrieve somewhat more complex information through other APIs.

2 APIs 101

APIs intends to serve as an intermediary between a client and a server. This client can be of two types: a web interface or programming software. The API makes no assumptions about the tool sending it a command; it simply requires adherence to a standard (usually an HTTP request), a query structure (the arguments), and then awaits the result.

2.1 Understanding with an interactive example

The first mode (access via a browser) is primarily used when a web interface allows a user to make choices in order to return results corresponding to those selections. Let's revisit the example of the geolocation API that we will use in this chapter. Imagine a web interface that offers the user two choices: a postal code and an address. These inputs will be injected into the query, and the server will respond with the appropriate geolocation.

2.2 How to Do It with `Python`?

The principle is the same, although we lose the interactive aspect. With `Python`, the idea is to construct the desired URL and fetch the result through an HTTP request.

We have already seen in the web scraping chapter how `Python` communicates with the internet via the `requests` package. This package follows the HTTP protocol where two main types of requests can be found: `GET` and `POST`:

- The `GET` request is used to retrieve data from a web server. It is the simplest and most common method for accessing the resources on a web page. We will start by describing this one.
- The `POST` request is used to send data to the server, often with the goal of creating or updating a resource. On web pages, it is commonly used for submitting forms that need to update information in a database (passwords, customer data, etc.). We will see its usefulness later, when we begin to deal with authenticated requests where additional information must be submitted with our query.

Let's conduct a first test with `Python` as if we were already familiar with this API.

```
import requests
adresse = "88 avenue verdier"
url_ban_example = f"https://data.geopf.fr/geocodage/search/?q={adresse.replace(" ", "+")}
&postcode=92120"
requests.get(url_ban_example)
```

```
<Response [200]>
```

What do we get? An HTTP code. The code 200 corresponds to successful requests, meaning that the server is able to respond. If this is not the case, for some reason *x* or *y*, you will receive a different code.

> HTTP Status Codes
>
> HTTP status codes are standard responses sent by web servers to indicate the result of a request made by a client (such as a web browser or a Python script). They are categorized based on the first digit of the code:
>
> - 1xx: Informational
> - 2xx: Success

- 3xx: Redirection
- 4xx: Client-side Errors
- 5xx: Server-side Errors

The key codes to remember are: 200 (success), 400 (bad request), 401 (authentication failed), 403 (forbidden), 404 (resource not found), 503 (the server is unable to respond)

To retrieve the content returned by `requests`, there are several methods available. When the JSON is well-formatted, the simplest approach is to use the `json` method, which converts it into a dictionary:

```
req = requests.get(url_ban_example)
localisation_insee = req.json()
localisation_insee
```

```
{'type': 'FeatureCollection',
 'features': [{'type': 'Feature',
   'geometry': {'type': 'Point', 'coordinates': [2.309144, 48.81622]},
   'properties': {'label': '88 Avenue Verdier 92120 Montrouge',
    'score': 0.9734354545454544,
    'housenumber': '88',
    'id': '92049_9625_00088',
    'banId': '92dd3c4a-6703-423d-bf09-fc0412fb4f89',
    'name': '88 Avenue Verdier',
    'postcode': '92120',
    'citycode': '92049',
    'x': 649270.67,
    'y': 6857572.24,
    'city': 'Montrouge',
    'context': '92, Hauts-de-Seine, Île-de-France',
    'type': 'housenumber',
    'importance': 0.70779,
    'depcode': '92',
    'street': 'Avenue Verdier',
    '_type': 'address'}}],
 'query': '88 avenue verdier'}
```

In this case, we can see that the data is nested within a JSON. Therefore, a bit of code needs to be written to extract the desired information from it:

```
localisation_insee.get('features')[0].get('properties')
```

```
{'label': '88 Avenue Verdier 92120 Montrouge',
 'score': 0.9734354545454544,
 'housenumber': '88',
 'id': '92049_9625_00088',
 'banId': '92dd3c4a-6703-423d-bf09-fc0412fb4f89',
 'name': '88 Avenue Verdier',
 'postcode': '92120',
 'citycode': '92049',
 'x': 649270.67,
 'y': 6857572.24,
 'city': 'Montrouge',
 'context': '92, Hauts-de-Seine, Île-de-France',
 'type': 'housenumber',
 'importance': 0.70779,
 'depcode': '92',
```

```
 'street': 'Avenue Verdier',
 '_type': 'address'}
```

This is the main disadvantage of using APIs: the post-processing of the returned data. The necessary code is specific to each API, since the structure of the JSON depends on the API.

2.3 How to Know the *Inputs* and *Outputs* of APIs?

Here, we took the BAN API as a magical tool whose main *inputs* (the endpoint, parameters, and their formatting…) were known. But how does one actually get there in practice? Simply by reading the documentation when it exists and testing it with examples.

Good APIs provide an interactive tool called `swagger`. It is an interactive website where the API's main features are described and where the user can interactively test examples. These documentations are often automatically created during the construction of an API and made available via an entry point `/docs`. They often allow you to edit certain parameters in the browser, view the obtained JSON (or the generated error), and retrieve the formatted query that produced it. These interactive browser consoles replicate the experimentation that can otherwise be done using specialized tools like `postman`.

Regarding the BAN API, the documentation can be found at https://www.data.gouv.fr/dataservices/api-adresse-base-adresse-nationale-ban/ (French only for now but you can translate it using DeepL or Google Translate). Go to "search" and click on "Try it out!" to toy with the interactive swagger. It provides many examples that can be directly tested from the browser, gives you the URL of the request, if the response is successful or not and the response content. You can also use the URLs provided as examples using `curl` (a command-line equivalent of `requests` in Linux):

```
curl "https://data.geopf.fr/geocodage/search/?q=8+bd+du+port&limit=15"
```

Just copy the URL (`https://data.geopf.fr/geocodage/search/?q=8+bd+du+port&limit=15`), open a new tab, and verify that it produces a result. Then change a parameter and check again until you find the structure that fits. After that, you can move on to `Python` as suggested in the following exercise.

2.4 Application

To start this exercise, you will need the following variable:

```
adresse = "88 Avenue Verdier"
```

> Exercise 1: Structure an API Call from Python
>
> 1. Test the API without any additional parameters, and convert the result into a `DataFrame`.
> 2. Limit the search to Montrouge using the appropriate parameter and find the corresponding INSEE code or postal code via Google.
> 3. (Optional): Display the found address on a map.

The first two rows of the `DataFrame` obtained in question 1 should be

	label	score	housenumber	id	banId	name	postcode	citycode	x	y	city	context	type	importance	depcode	street	_type	locality
0	88 Avenue Verdier 92120 Montrouge	0.973435	88	92049_9625_00088	92dd3c4a-6703-423d-bf09-fc0412fb4f89	88 Avenue Verdier	92120	92049	649270.67	6857572.24	Montrouge	92, Hauts-de-Seine, Île-de-France	housenumber	0.70779	92	Avenue Verdier	address	NaN
1	Avenue Verdier 44500 La Baule-Escoublac	0.719494	NaN	44055_3690	e6e5569-c6ba-49fc-bbf8-863301a8b2df	Avenue Verdier	44500	44055	291895.65	6701236.88	La Baule-Escoublac	44, Loire-Atlantique, Pays de la Loire	street	0.60193	44	Avenue Verdier	address	NaN

For question 2, this time we get back only one observation, which could be further processed with `GeoPandas` to verify that the point has been correctly placed on a map.

	label	score	housenumber	id	banId	name	postcode	citycode	x	y	city	context	type	importance	depcode	street	_type	geometry
0	88 Avenue Verdier 92120 Montrouge	0.973435	88	92049_9625_00088	92dd3c4a-6703-423d-bf09-fc0412fb4f89	88 Avenue Verdier	92120	92049	649270.67	6857572.24	Montrouge	92, Hauts-de-Seine, Île-de-France	housenumber	0.70779	92	Avenue Verdier	address	POINT (2.30914 48.81622)

Finally, for question 3, we obtain this map (more or less the same as before):

3 More Examples of `GET` Requests

3.1 Main Source

We will use as the main basis for this tutorial the permanent equipment database, a directory of public facilities open to the public.

We will begin by retrieving the data that interest us. Rather than fetching every variable in the file, we only retrieve the ones we need: some variables concerning the facility, its address, and its local municipality.

We will restrict our scope to primary, secondary, and higher education institutions in the department of Haute-Garonne (department 31). These facilities are identified by a specific code, ranging from `C1` to `C5`.

```
import duckdb

query = """
FROM read_parquet('https://minio.lab.sspcloud.fr/lgaliana/diffusion/BPE23.parquet')
SELECT NOMRS, NUMVOIE, INDREP, TYPVOIE, LIBVOIE,
       CADR, CODPOS, DEPCOM, DEP, TYPEQU,
       concat_ws(' ', NUMVOIE, INDREP, TYPVOIE, LIBVOIE) AS adresse, SIRET
WHERE DEP = '31'
      AND NOT (starts_with(TYPEQU, 'C6') OR starts_with(TYPEQU, 'C7'))
"""

bpe = duckdb.sql(query)
bpe = bpe.to_df()
bpe = bpe[bpe['TYPEQU'].str.startswith('C')]
```

```
FloatProgress(value=0.0, layout=Layout(width='auto'), style=ProgressStyle(bar_color='black'))
```

3.2 Retrieving Custom Data via APIs

We previously covered the general principle of an API request. To further illustrate how to retrieve data on a larger scale using an API, let's try to fetch supplementary data to our main source. We will use the education directory, which provides extensive information on educational institutions. We will use the SIRET number to cross-reference the two data sources.

The following exercise will demonstrate the advantage of using an API to obtain custom data and the ease of fetching it via `Python`. However, this exercise will also highlight one of the limitations of certain APIs, namely the volume of data that needs to be retrieved.

Exercise 2

1. Visit the *swagger* of the National Education Directory API on data.gouv.fr and test an initial data retrieval using the `records` endpoint without any parameters.

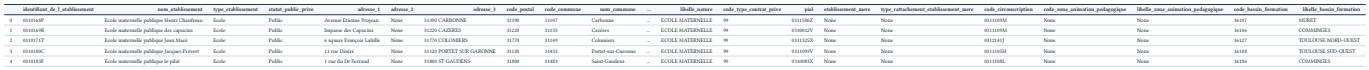
2. Since we have retained only data from Haute Garonne in our main database, we want to retrieve only the institutions from that department using our API. Make a query with the appropriate parameter, without adding any extras.
3. Increase the limit on the number of parameters—do you see the problem?
4. API metadata contains the total count of rows. By looking thoroughly at the API documentation, do you see a way to get the whole dataset?

The first question allows us to retrieve an initial dataset.



identifiant_de_l_etablissement	nom_etablissement	type_etablissement	statut_public_prive	adresse_1	adresse_2	adresse_3	code_postal	code_commune	nom_commune	...	libelle_nature	code_type_contrat_prive	pial	etablissement_mere	type_rattachement_etablissement_mere	code_circonscription	code_zone_animation_pedagogique	libelle_zone_animation_pedagogique	code_bassin_formation	libelle_bassin_formation
0 0181165E	Ecole primaire privée Les Marronniers	Ecole	Privé	95 rue des Frères	BP 67	38630 CORBELIN	38630	38124	Corbelin	...	ECOLE DE NIVEAU ELEMENTAIRE	10	0180044U	None	None	0182753N	None	None	0180	NOMD ISERE
1 0182671X	Ecole primaire La Plaine	Ecole	Public	12 rue des Quatre Saisons	MaN	38230 TIGNIEU JAMEYZIEU	38230	38545	Tignieu-Jameyzieu	...	ECOLE DE NIVEAU ELEMENTAIRE	99	0180011F	None	None	0181623H	None	None	0180	NOMD ISERE

However, there are two issues: there are only 10 rows and the datasets covers the whole French territory when we want to restrict to our department of interest. Let's first address the latter with question 2.



	identifiant_de_l_etablissement	nom_etablissement	type_etablissement	statut_public_prive	adresse_1	adresse_2	adresse_3	code_postal	code_commune	nom_commune	...	libelle_nature	code_type_contrat_prive	pial	etablissement_mere	type_rattachement_etablissement_mere	code_circonscription	code_zone_animation_pedagogique	libelle_zone_animation_pedagogique	code_bassin_formation	libelle_bassin_formation
0	0310164P	Ecole maternelle publique Henri Chatibton	Ecole	Public	Avenue Etienne Poujeau	None	31390 CARBONNE	31390	31097	Carbonne	...	ECOLE MATERNELLE	99	0311186Z	None	None	0311109M	None	None	16107	MURET
1	0310164R	Ecole maternelle publique des capucins	Ecole	Public	Impasse des Capucins	None	31220 CAZERES	31220	31135	Cazères	...	ECOLE MATERNELLE	99	0310012V	None	None	0311109M	None	None	16106	COMMINGES
2	0310171T	Ecole maternelle publique Jean Macé	Ecole	Public	6 square François Lahille	None	31770 COLOMIERS	31770	31149	Colomiers	...	ECOLE MATERNELLE	99	0311102X	None	None	0312141J	None	None	16127	TOULOUSE NORD-OUEST
3	0310183C	Ecole maternelle publique Jacques Prévert	Ecole	Public	11 rue Daizé	None	31120 PORTET SUR GARONNE	31120	31455	Portet-sur-Garonne	...	ECOLE MATERNELLE	99	0311049Y	None	None	0311165H	None	None	16103	TOULOUSE SUD-OUEST
4	0310181F	Ecole maternelle publique le pilat	Ecole	Public	1 rue du Dr Ferrand	None	31860 ST GAUDENS	31860	31483	Saint-Gaudens	...	ECOLE MATERNELLE	99	0310003X	None	None	0311108L	None	None	16106	COMMINGES

This is better, but we still only have 10 observations. If we try to adjust the number of rows (question 3), we get the following response from the API:

```
b'{\n  "error_code": "InvalidRESTParameterError",\n  "message": "Invalid value for limit API
parameter: 200 was found but -1 ⩽ limit ⩽ 100 is expected."\n}'
```

By looking at the data metadata, we notice that a variable `total_count` gives us the dataset size (1269 rows in our case). We will use the `offset` field to get data from the nth row and download the whole dataset. Since the automation code is rather tedious to write, here it is:

```
import requests
import pandas as pd

# Initialize the initial API URL
dep = '031'
offset = 0
limit = 100
url_api_datagouv   =   f'https://data.education.gouv.fr/api/v2/catalog/datasets/fr-en-annuaire-
education/records?where=code_departement=\'{dep}\'&limit={limit}&offset={offset}'

# First request to get total obs and first df
response = requests.get(url_api_datagouv)
nb_obs = response.json()['total_count']
schools_dep31 = pd.json_normalize(response.json()['records'])

# Loop on the nb of obs
while nb_obs > len(schools_dep31):
    try:
        # Increase offset for first reply sent and update API url
        offset += limit
            url_api_datagouv  =  f'https://data.education.gouv.fr/api/v2/catalog/datasets/fr-en-
annuaire-education/records?where=code_departement=\'{dep}\'&limit={limit}&offset={offset}'

        # Call API
        print(f'fetching from {offset}th row')
        response = requests.get(url_api_datagouv)

        # Concatenate the data from this call to previous data
        page_data = pd.json_normalize(response.json()['records'])
        schools_dep31 = pd.concat([schools_dep31, page_data], ignore_index=True)
```

```
        print(f'length of data is now {len(schools_dep31)}')

    except requests.exceptions.RequestException as e:
        print(f"An error occurred: {e}")
        break

# Filter columns to keep only those starting with 'record.fields'
schools_dep31 = schools_dep31.filter(regex='^record\\.fields\\..*')

# Rename columns by removing 'record.fields.' at the beginning
schools_dep31 = schools_dep31.rename(columns=lambda x: x.replace('record.fields.', ''))
```

The resulting *DataFrame* is as follows:

```
schools_dep31.head()
```

	identifiant_de_l_etablissement	nom_etablissement	type_etablissement	statut_public_prive	adresse_1	adresse_2	adresse_3	code_postal	code_commune	nom_commune	...	code_type_contrat_prive	pial	etablissement_mere	type_rattachement_etablissement_mere	code_circonscription	code_zone_animation_pedagogique	libelle_zone_animation_pedagogique	code_bassin_formation	libelle_bassin_formation	position
0	0310166P	Ecole maternelle publique Henri Chanfreau	Ecole	Public	Avenue Etienne Precjean	None	31390 CARBONNE	31390	31107	Carbonne	...	99	0311538Z	None	None	0311109M	None	None	16107	MURET	NaN
1	0310169R	Ecole maternelle publique des capucins	Ecole	Public	Impasse des Capucins	None	31220 CAZERES	31220	31135	Cazères	...	99	0310012V	None	None	0311109M	None	None	16106	COMMINGES	NaN
2	0310171T	Ecole maternelle publique Jean Macé	Ecole	Public	6 square François Labilla	None	31770 COLOMIERS	31770	31149	Colomiers	...	99	0311121X	None	None	0312114Q	None	None	16127	TOULOUSE NORD-OUEST	NaN
3	0310180C	Ecole maternelle publique Jacques Prévert	Ecole	Public	11 rue Diviné	None	31120 PORTET SUR GARONNE	31120	31431	Portet-sur-Garonne	...	99	0311095V	None	None	0311105H	None	None	16108	TOULOUSE SUD-OUEST	NaN
4	0310181F	Ecole maternelle publique le pilat	Ecole	Public	1 rue du Dr Ferrand	None	31800 ST GAUDENS	31800	31483	Saint-Gaudens	...	99	0310003X	None	None	0311110L	None	None	16106	COMMINGES	NaN

We can merge this new data with our previous dataset to enrich it. For reliable production, care should be taken with schools that do not match, but this is not critical for this series of exercises.

```
bpe_enriched = bpe.merge(
  schools_dep31,
  left_on = "SIRET",
  right_on = "siren_siret"
)
bpe_enriched.head(2)
```

	NOMRS	NUMVOIE	INDREP	TYPVOIE	LIBVOIE	CADR	CODPOS	DEPCOM	DEP	TYPEQU	...	code_type_contrat_prive	pial	etablissement_mere	type_rattachement_etablissement_mere	code_circonscription	code_zone_animation_pedagogique	libelle_zone_animation_pedagogique	code_bassin_formation	libelle_bassin_formation	position
0	ECOLE PRIMAIRE PUBLIQUE DENIS LATAPIE			LD	LA BOURDETTE		31230	31001	31	C108	...	99	0310003X	NaN	NaN	0311108L	None	None	16106	COMMINGES	NaN
1	ECOLE MATERNELLE PUBLIQUE	21		CHE	DE L AUTAN		31280	31003	31	C107	...	99	0311335H	NaN	NaN	0311102E	None	None	16128	TOULOUSE EST	NaN

This provides us with data enriched with new characteristics about the institutions. Although there are geographic coordinates in the dataset, we will pretend there aren't to reuse our geolocation API.

4 Discovering `POST` Requests

4.1 Logic

So far, we have discussed `GET` requests. Now, we will introduce `POST` requests, which allow for more complex interactions with API servers.

To explore this, we will revisit the previous geolocation API but use a different endpoint that requires a `POST` request.

`POST` requests are typically used when specific data needs to be sent to trigger an action. For instance, in the web world, if authentication is required, a `POST` request can send a token to the server, which will respond by accepting your authentication.

In our case, we will send data to the server, which will process it for geolocation and then send us a response. To continue the culinary metaphor, it's like handing over your own container (*tupperware*) to the kitchen to collect your takeaway meal.

4.2 Principle

Let's look at this request provided on the geolocation API's documentation site:

```
curl -X POST -F data=@path/to/file.csv -F columns=voie -F columns=ville -F
citycode=ma_colonne_code_insee https://data.geopf.fr/geocodage/search/csv
```

As mentioned earlier, `curl` is a command-line tool for making API requests. The `-X POST` option clearly indicates that we want to make a `POST` request.

Other arguments are passed using the `-F` options. In this case, we are sending a file and adding parameters to help the server locate the data inside it. The `@` symbol indicates that `file.csv` should be read from the disk and sent in the request body as form data.

4.3 Application with `Python`

We have `requests.get`, so naturally, we also have `requests.post`. This time, parameters must be passed to our request as a dictionary, where the keys are argument names and the values are `Python` objects.

The main challenge, illustrated in the next exercise, lies in passing the `data` argument: the file must be sent as a `Python` object using the `open` function.

Exercise 3: A POST request to geolocate our data in bulk

1. Save the `adresse`, `DEPCOM`, and `nom_commune` columns of the equipment database merged with our previous directory (object `bpe_enriched`) in CSV format. Before writing to CSV, it may be helpful to replace commas in the `adresse` column with spaces.
2. Create the `response` object using `requests.post` with the correct arguments to geocode your CSV.
3. Transform your output into a `geopandas` object using the following command:

```
bpe_loc = pd.read_csv(io.StringIO(response.text))
```

The obtained geolocations take this form

	index	adresse	DEPCOM	nom_commune	result_score	latitude	longitude
0	0	LD LA BOURDETTE	31001	Agassac	0.404457	43.374288	0.880679
1	1	21 CHE DE L AUTAN	31003	Aigrefeuille	0.730598	43.567370	1.585932

By enriching the previous data, this gives:

	NOMRS	NUMVOIE	INDREP	TYPVOIE	LIBVOIE	CADR	CODPOS	DEPCOM	DEP	TYPEQU	...	type_rattachement_etablissement_mere	code_circonscription	code_zone_animation_pedagogique	libelle_zone_animation_pedagogique	code_bassin_formation	libelle_bassin_formation	position	result_score	latitude_ban	longitude_ban
0	ECOLE PRIMAIRE PUBLIQUE DENIS LATAPIE			LD	LA BOURDETTE		31250	31001	31	C108	...	NaN	0311108L	None	None	16106	COMMINGES	NaN	0.404457	43.374288	0.880679
1	ECOLE MATERNELLE PUBLIQUE	21		CHE	DE L AUTAN		31260	31003	31	C107	...	NaN	0311102E	None	None	16128	TOULOUSE EST	NaN	0.730598	43.567370	1.585932

We can check that the geolocation is not too off by comparing it with the longitudes and latitudes of the education directory added earlier:

	NOMRS	nom_commune	longitude_annuaire	longitude_ban	latitude_annuaire	latitude_ban
671	ECOLE ELEMENTAIRE PUBLIQUE LES TROIS CHENES	La Salvetat-Saint-Gilles	1.260430	1.260305	43.580879	43.581660
398	ECOLE PRIMAIRE PUBLIQUE	La Magdelaine-sur-Tarn	1.544200	1.543327	43.810691	43.811190
302	ECOLE PRIMAIRE PUBLIQUE FRANCIS SANCERRY	Gratens	1.111270	1.123272	43.322120	43.325913
752	ECOLE MATERNELLE PUBLIQUE DIDIER DAURAT	Toulouse	1.402211	1.402215	43.573382	43.573382
1043	ECOLE ELEMENTAIRE PUBLIQUE COMTESSE DE SEGUR	Verfeil	1.663350	1.663512	43.659332	43.659041

Without going into detail, the positions seem very similar, with only minor inaccuracies.

To make use of our enriched data, we can create a map. To add some context to it, we can place a background map of the municipalities. This can be retrieved using `cartiflette`:

```
from cartiflette import carti_download
shp_communes = carti_download(
  crs = 4326,
  values = ["31"],
  borders="COMMUNE",
  filter_by="DEPARTEMENT",
  source="EXPRESS-COG-CARTO-TERRITOIRE",
  year=2022
)
shp_communes.crs = 4326
```

```
/__w/python-datascientist/python-datascientist/.venv/lib/python3.13/site-packages/pandas/core/
generic.py:6208: FutureWarning: Overriding the CRS of a GeoDataFrame that already has CRS. This
unsafe behavior will be deprecated in future versions. Use GeoDataFrame.set_crs method instead
  return object.__setattr__(self, name, value)
```

Represented on a map, this gives the following map:

5 Managing secrets and exceptions

We have already used several APIs. However, these APIs were all without authentication and had few restrictions, except for the number of calls. This is not the case for all APIs. It is common for APIs that allow access to more data or confidential information to require authentication to track data users.

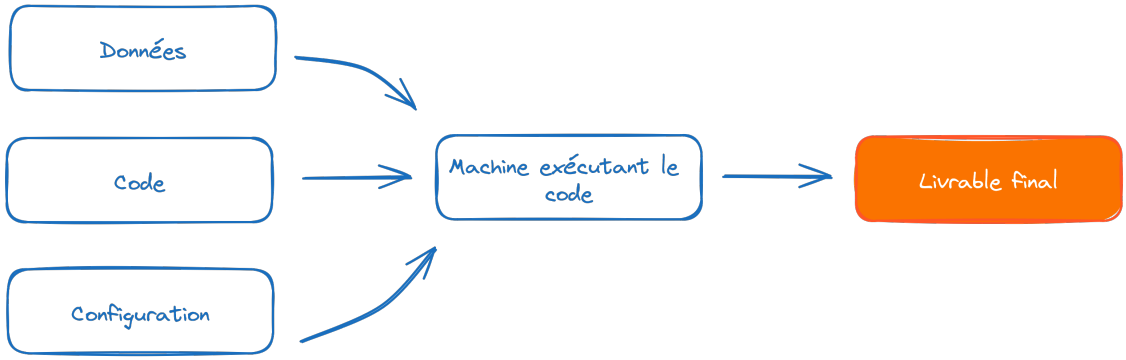
This is usually done through a *token*. A token is a kind of password often used in modern authentication systems to certify a user's identity (see Git chapter).

To illustrate the use of *tokens*, we will use an API from Insee, French national statistical institute. We will use this API to retrieve entreprise level data from an exhaustive registry.

Before diving into this, we will take a detour to discuss token confidentiality and how to avoid exposing tokens in your code.

5.1 How to use a token in code without revealing it

Tokens are personal information that should not be shared. There values are not meant to be present in the code. As mentioned multiple times in the production deployment course taught by Romain Avouac and myself in the third year, it is crucial to separate the code from configuration elements.



The idea is to find a way to include configuration elements with the code without exposing them directly in the code. The general approach is to store the token value in a variable without revealing it in the code. How can we declare the token value without making it visible in the code?

- For interactive code (e.g., via a *notebook*), it is possible to create a dialog box that injects the provided value into a variable. This can be done using the `getpass` package.
- For non-interactive code, such as command-line scripts, the environment variable approach is the most reliable, provided you are careful not to include the password file in `Git`.

The following exercise will demonstrate these two methods. These methods will help us confidentially add a *payload* to authentication requests, i.e., confidential identifying information as part of a request.

5.2 Understanding the principle with interactive documentation

To illustrate the use of authenticated APIs, we suggest exploring Insee's API portal.

We will focus on the Sirene API but, on this portal, there are others, notably the Melodi API dedicated to retrieving a number of Insee *open data* sources. The Sirene API is a queryable version of the Sirene *open data* (in French).

Before going to `Python`, let us use our browser to understand the principle of authentication.

Exercise 4: API authenticated through the browser

1. Create an account on Insee API portal (seems to be only in French, sorry about that).
2. Go to the `My Applications` section and create a new one. For simplicity, you can name it `Python data science`. While creating this application, subscribe to the Sirene API.
3. Click on the Subscriptions tab. Select the Sirene API, which should normally be available at the bottom of the page.
4. On the right, you should now see a token displayed. Keep this page open and open portail-api.insee.fr/catalog/all in a new tab.

Now let's test the Sirene API using *swagger*, starting from portail-api.insee.fr/catalog/all.

1. Click on the `Documentation` tab. The interactive documentation (the *swagger*) should now be displayed.
2. Further down in the interactive documentation, go to the entry point `/siren/{siren}` (`GET` method).
3. Click on the `Try it out` button to try out the example interactively.
4. Fill in the `siren` field with the SIREN `500569405`, which is Decathlon's identifier 😉 If you get the error below, do you understand why?

```
//| echo: false
error_request_no_token
```

You will need to be authenticated.

1. To do this, reload this interactive documentation and click on the `Authorise` button. In the pop-up that appears, enter the token from the other window.
2. After validating this token, try the previous procedure for retrieving information from Decathlon again.

You should now see this output:

```
//| echo: false
sortie_sirene_decathlon
```

5.3 Application

For this application, starting from question 4, we will need to create a special class that allows `requests` to override our request with an authentication token. Since it is not trivial to create without prior knowledge, here it is:

```
class BearerAuth(requests.auth.AuthBase):
    def __init__(self, token):
        self.token = token
    def __call__(self, r):
        r.headers["X-INSEE-Api-Key-Integration"] = self.token
        return r
```

```
siren = "500569405"
```

Exercice 5 : API tokens with Python

1. Create a `token` variable using `getpass`.
2. Use this code structure to retrieve the desired data.

```
requests.get(
    url,
    auth=BearerAuth(token)
)
```

3. Replace `getpass` snippet with the environment variable approach using `dotenv`.

Important

The environment variable approach is the most general and flexible. However, it is crucial to ensure that the `.env` file storing the credentials is not added to `Git`. Otherwise, you risk exposing identifying information, which negates any benefits of the good practices implemented with `dotenv`.

The solution is simple: add the `.env` line to `.gitignore` and, for extra safety, include `*.env` in case the file is not at the root of the repository. To learn more about the `.gitignore` file, refer to the Git chapters.

6 Opening Up to Model APIs

So far, we have explored data APIs, which allow us to retrieve data. However, this is not the only interesting use case for APIs among `Python` users.

There are many other types of APIs, and model APIs are particularly noteworthy. They allow access to pre-trained models or even perform inference on specialized servers with more resources than a local computer (more details in the *machine learning* and NLP sections). The most well-known library in this field is the `transformers` library developed by `HuggingFace`.

One of the objectives of the 3rd-year production deployment course is to demonstrate how this type of software architecture works and how it can be implemented for models you have created yourself.

7 Additional Exercises: let's add the high schools added value

> Bonus Exercise
>
> In our example on schools, limit the scope to high schools and add information on the added value of high schools available here.

> Bonus Exercise 2: Where are we going out tonight?
>
> Finding a common place to meet friends is always a subject of tough negotiations. What if we let geography guide us?
>
> 1. Create a `DataFrame` recording a series of addresses and postal codes, like the example below.
> 2. Adapt the code from the exercise on the BAN API, using its documentation, to geolocate these addresses.
> 3. Assuming your geolocated data is named `adresses_geocoded`, use the proposed code to transform them into a polygon.
> 4. Calculate the centroid and display it on an interactive `Folium` map as before.
>
> You forgot there's a couple in the group... Take into account the `poids` variable to calculate the barycenter and find out where to meet tonight.
>
> Create the polygon from the geolocations
>
> ```
> from shapely.geom
> etry import Polygon
> coordinates = list(zip(adresses_geocoded['longitude'], adresses_geocoded['latitude']))
> polygon = Polygon(coordinates)
>
> polygon = gpd.GeoDataFrame(index=[0], crs='epsg:4326', geometry=[polygon])
> polygon
> ```

The example DataFrame:

```
adresses_text = pd.DataFrame(
  {
    "adresse": [
        "10 Rue de Rivoli",
        "15 Boulevard Saint-Michel",
        "8 Rue Saint-Honoré",
        "20 Avenue des Champs-Élysées",
        "Place de la Bastille",
```

```
    ],
    "cp": ["75004", "75005", "75001", "75008", "75011"],
    "poids": [2, 1, 1, 1, 1]
})
adresses_text
```

	adresse	cp	poids
0	10 Rue de Rivoli	75004	2
1	15 Boulevard Saint-Michel	75005	1
2	8 Rue Saint-Honoré	75001	1
3	20 Avenue des Champs-Élysées	75008	1
4	Place de la Bastille	75011	1

	adresse	poids	cp	result_score	latitude	longitude
0	10 Rue de Rivoli	2	75004	0.970158	48.855500	2.360410
1	15 Boulevard Saint-Michel	1	75005	0.973296	48.851852	2.343614
2	8 Rue Saint-Honoré	1	75001	0.866200	48.863801	2.335725
3	20 Avenue des Champs-Élysées	1	75008	0.872216	48.871049	2.303730
4	Place de la Bastille	1	75011	0.965214	48.853711	2.370213

The geolocation obtained for this example

Here is the map obtained from the example dataset. We might have a better location with the barycenter than with the centroid.

```
apiroot = "https://data.geopf.fr/geocodage"
param1 = {
  const AdresseFormat = adresse.toLowerCase().replaceAll(" ", "+")
  const url = `q=${AdresseFormat}`
  return url
}
param2 = `postcode=${codePostal}`
```

```
import {mj} from "@danielefadda/mathjax"
```

```
url = {
  const AdresseFormat = adresse.toLowerCase().replaceAll(" ", "+")
 const url = `https://data.geopf.fr/geocodage/search/?q=${AdresseFormat}&postcode=${codePostal}
`
  return url
}
```

```
localisation = d3.json(url)
```

```
defaultAdresse = "88 Avenue Verdier"
longitude = localisation.features[0].geometry.coordinates[0]
latitude = localisation.features[0].geometry.coordinates[1]
```

```
map = {
  const container = html`<div style="height:300px;">`;
```

```
  yield container;
  const map = L.map(container).setView([latitude, longitude], 13);
  L.tileLayer("https://{s}.tile.openstreetmap.org/{z}/{x}/{y}.png", {
      attribution: "&copy; <a href=https://www.openstreetmap.org/copyright>OpenStreetMap</a>
contributors"
  }).addTo(map);
  var marker = L.marker([latitude, longitude]).addTo(map);
  marker.bindPopup("<b>Trouvé !</b>").openPopup();
  return map
}
```

```
import {L} from "@observablehq/hello-leaflet"
```

```
<IPython.core.display.HTML object>
```

Informations additionnelles

i Python environment

This site was built automatically through a `Github` action using the `Quarto` reproducible publishing software (version 1.10.18).

The environment used to obtain the results is reproducible via uv. The `pyproject.toml` file used to build this environment is available on the `linogaliana/python-datascientist` repository

```
[project]
name = "python-datascientist"
version = "0.1.0"
description = "Source code for Lino Galiana's Python for data science course"
readme = "README.md"
requires-python = "≥3.13,<3.14"
dependencies = [
    "altair≥6.0.0",
    "cartiflette≥0.2",
    "contextily==1.6.2",
    "duckdb≥0.10.1",
    "folium≥0.19.6",
    "gdal==3.11.4",
    "graphviz==0.20.3",
    "great-tables≥0.12.0",
    "gt-extras≥0.0.8",
    "ipykernel≥6.29.5",
    "jupyter≥1.1.1",
    "jupyter-cache≥1.0.0",
    "kaleido≥0.2.1",
    "langchain-community≥0.3.27",
    "loguru==0.7.3",
    "markdown≥3.8",
    "nbclient≥0.10.0",
    "nbformat≥5.10.4",
    "nltk≥3.9.1",
    "openpyxl≥3.1.5",
    "pandas≥3.0",
```

```
    "pip≥25.1.1",
    "plotly≥6.1.2",
    "plotnine≥0.15",
    "polars≥1.8.2",
    "py7zr≥1.1.3",
    "pyarrow≥17.0.0",
    "pynsee≥0.1.8",
    "python-dotenv≥1.0.1",
    "python-frontmatter≥1.1.0",
    "pywaffle≥1.1.1",
    "requests≥2.32.3",
    "scikit-image≥0.24.0",
    "scikit-learn≥1.8.0",
    "scipy≥1.13.0",
    "seaborn≥0.13.2",
    "selenium<4.39.0",
    "spacy≥3.8.4",
    "webdriver-manager≥4.0.2",
    "wordcloud==1.9.3",
]

[tool.uv.sources]
gdal = [
  { index = "gdal-wheels", marker = "sys_platform == 'linux'" },
  { index = "geospatial_wheels", marker = "sys_platform == 'win32'" },
]

[[tool.uv.index]]
name = "geospatial_wheels"
url = "https://nathanjmcdougall.github.io/geospatial-wheels-index/"
explicit = true

[[tool.uv.index]]
name = "gdal-wheels"
url = "https://gitlab.com/api/v4/projects/61637378/packages/pypi/simple"
explicit = true

[dependency-groups]
dev = [
    "nb-clean≥4.0.1",
]
```

To use exactly the same environment (version of `Python` and *packages*), please refer to the documentation for `uv`.